

D471

Global Supply Chain Management

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A toy maker sees consumer sales vary about 5% month to month, but orders from its distributors swing 40%. Its headquarters is in a converted mill and its best seller is a wooden train. What most directly reduces the swings?

- A. Order in larger batches to get volume discounts.
- B. Raise safety stock at every tier.
- C. Nothing can be done; consumer demand is the cause.
- D. Share point-of-sale data across tiers and cut lead times and batch sizes.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Under the bullwhip effect, where in a supply chain is order variability usually greatest?

- A. At raw-material suppliers, farthest upstream.
- B. Equal at every tier.
- C. At the retailer.
- D. At the end consumer.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A bicycle company, which sponsors a racing team, builds frames to forecast but paints and assembles each bike only after a customer picks colors online. Where is the push-pull boundary?

- A. At the customer; the whole chain is push.
- B. At raw materials; the whole chain is pull.
- C. At the racing team's bike shop.
- D. Between frame building and painting.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A medical device firm makes a component in-house at a variable cost of \$18 per unit plus \$200,000 a year in fixed cost that would disappear if it outsourced. A supplier quotes \$22 per unit, and managing the supplier would cost \$30,000 a year. Annual volume is 60,000 units. The supplier's CEO once worked at the firm. Which option is cheaper, and by how much?

- A. Make, by \$240,000.
- B. Buy, by \$70,000.
- C. Make, by \$70,000.
- D. Make, by \$40,000.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Making costs \$18 per unit plus \$200,000 in avoidable fixed cost; buying costs \$22 per unit plus \$30,000 to manage the supplier. At what annual volume are the two equal?

- A. 42,500 units
- B. 57,500 units
- C. 50,000 units
- D. About 7,727 units

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

An electronics assembler rates two risks. Risk A, a port strike: 10% annual probability, \$4 million impact. Risk B, failure of its sole-source chip supplier: 5% probability, \$12 million impact. The supply chain VP has 20 years at the company. Which risk has the higher expected loss?

- A. Neither; low-probability risks can be ignored.
- B. Risk B: \$600,000 vs. \$400,000.
- C. Risk B, with an expected loss of \$12 million.
- D. Risk A, because its probability is twice as high.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A risk has a \$2 million impact. Leaders will accept an expected loss of up to \$150,000. What is the highest annual probability they will accept?

- A.** 0%; no risk is acceptable.
- B.** 15%
- C.** 7.5%
- D.** About 13.3%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

At a consumer goods firm, sales forecasts 1.2 million cases next quarter, operations plans 950,000 based on plant capacity, and finance budgets from a third number. The company recently opened an innovation lab. What does a mature S&OP process do?

- A.** Has operations build to the 1.2 million sales forecast.
- B.** Lets each department keep its own plan.
- C.** Holds a monthly cross-functional review that reconciles demand, supply, and finance into one plan approved by executives.
- D.** Runs a weekly production scheduling meeting.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A hospital network buys a specialized imaging contrast agent from one of only two qualified suppliers worldwide; it is critical to patient care and a large share of spend. The procurement office was newly renovated. Which approach fits?

- A.** Frequent competitive bidding to push prices down.
- B.** A long-term strategic partnership with shared planning and risk management.
- C.** Stockpile it and look for substitutes, treating it as low spend.
- D.** Buy on purchasing cards with minimal oversight.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A distributor had cost of goods sold of \$48 million last year and average inventory of \$6 million. Sales were \$60 million. It moved into a new 300,000-square-foot facility. What are its inventory turns and days of supply (365-day year)?

- A.** 8 turns, about 46 days
- B.** 10 turns, about 37 days
- C.** 8 turns, about 8 days
- D.** 0.125 turns, about 2,920 days

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A retailer targets 30 days of supply (use a 360-day year) and has annual COGS of \$36 million. What average inventory does that imply?

- A.** \$1.08 billion
- B.** \$3.0 million
- C.** \$36 million
- D.** \$1.2 million

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A clothing brand with 400 stores and a new CEO wants to cut greenhouse gas emissions across its supply chain. Most emissions come from contract factories that dye and sew its garments. Under the GHG Protocol, those factory emissions are:

- A.** Not the brand's responsibility to measure
- B.** Scope 2
- C.** Scope 1
- D.** Scope 3

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A toy maker sees consumer sales vary about 5% month to month, but orders from its distributors swing 40%. ~~Its headquarters is in a converted mill and its best seller is a wooden train.~~ What most directly reduces the swings?

Target word: bullwhip remedy **Decides:** Upstream swings far exceed real demand swings.

A	Inverted operation	Batching is one of the causes of bullwhip.
B	Adjacent category	Absorbs swings at high cost without removing the cause.
C	Absolute language	Consumer demand is stable; the amplification is structural.
D	Correct	Removes the guesswork and batching that amplify orders.

2. Answer A Concept 1 • transfer

Under the bullwhip effect, where in a supply chain is order variability usually greatest?

Target word: reverse: where the effect peaks **Decides:** Amplification builds upstream.

A	Correct	Each tier amplifies the swings it receives.
B	Absolute language	The effect is defined by growth upstream.
C	Inverted operation	Retailers see the least amplified orders.
D	Confusion set	Consumer demand is the stable starting signal.

3. Answer D Concept 2

A bicycle company, ~~which sponsors a racing team,~~ builds frames to forecast but paints and assembles each bike only after a customer picks colors online. Where is the push-pull boundary?

Target word: decoupling point **Decides:** Frames are built ahead; painting waits for orders.

A	Inverted operation	Painting and assembly wait for orders.
B	Dropped element	Drops the forecast-driven frame stage.
C	Camouflage	The sponsorship does not locate the boundary.
D	Correct	Push before it, pull after it.

4. Answer C Concept 3

A medical device firm makes a component in-house at a variable cost of \$18 per unit plus \$200,000 a year in fixed cost that would disappear if it outsourced. A supplier quotes \$22 per unit, and managing the supplier would cost \$30,000 a year. Annual volume is 60,000 units. ~~The supplier's CEO once worked at the firm.~~ Which option is cheaper, and by how much?

Target word: cheaper option and gap **Decides:** Make = \$1,280,000; buy = \$1,350,000.

A	Adjacent category	Compares variable cost only, ignoring both fixed amounts.
B	Inverted operation	Right gap, wrong direction.
C	Correct	$18 \times 60,000 + 200,000 = 1,280,000$; $22 \times 60,000 + 30,000 = 1,350,000$.
D	Dropped element	Leaves out the \$30,000 supplier-management cost.

5. Answer A Concept 3 • transfer

Making costs \$18 per unit plus \$200,000 in avoidable fixed cost; buying costs \$22 per unit plus \$30,000 to manage the supplier. At what annual volume are the two equal?

Target word: solve for break-even volume **Decides:** $18x + 200,000 = 22x + 30,000$.

A	Correct	$170,000 / 4 = 42,500$.
B	Inverted operation	Adds the fixed costs (230,000 / 4) instead of taking the difference.
C	Dropped element	$200,000 / 4$ drops the \$30,000.
D	Wrong stem number	Divides by \$22 instead of the \$4 cost difference.

6. Answer B Concept 4

An electronics assembler rates two risks. Risk A, a port strike: 10% annual probability, \$4 million impact. Risk B, failure of its sole-source chip supplier: 5% probability, \$12 million impact. ~~The supply chain VP has 20 years at the company.~~ Which risk has the higher expected loss?

Target word: higher expected loss **Decides:** A: \$400,000; B: \$600,000.

A	Absolute language	Both carry six-figure expected losses.
B	Correct	$0.05 \times 12M = 600,000$; $0.10 \times 4M = 400,000$.
C	Wrong stem number	The impact, not the expected loss.
D	Dropped element	Ignores impact.

7. Answer C Concept 4 • transfer

A risk has a \$2 million impact. Leaders will accept an expected loss of up to \$150,000. What is the highest annual probability they will accept?

Target word: solve for probability **Decides:** $150,000 / 2,000,000$.

A	Absolute language	Leaders set a tolerance above zero.
B	Wrong stem number	Reads \$150,000 as a percentage.
C	Correct	$150,000 / 2,000,000 = 0.075$.
D	Inverted operation	$2,000,000 / 150,000$ read as a percent.

8. Answer C Concept 5

At a consumer goods firm, sales forecasts 1.2 million cases next quarter, operations plans 950,000 based on plant capacity, and finance budgets from a third number. ~~The company recently opened an innovation lab.~~ What does a mature S&OP process do?

Target word: S&OP role **Decides:** Three conflicting plans.

A	Dropped element	Drops the capacity limit.
B	Inverted operation	The opposite of alignment.
C	Correct	One agreed set of numbers is the purpose of S&OP.
D	Adjacent category	Scheduling is tactical execution, not S&OP.

9. Answer B Concept 6

A hospital network buys a specialized imaging contrast agent from one of only two qualified suppliers worldwide; it is critical to patient care and a large share of spend. ~~The procurement office was newly renovated.~~ Which approach fits?

Target word: sourcing approach **Decides:** High spend plus high supply risk = strategic.

A	Adjacent category	Suits leverage items with many suppliers.
B	Correct	Strategic items need close collaboration.
C	Dropped element	Bottleneck tactics drop the high-spend side.
D	Confusion set	Suits non-critical items.

10. Answer A Concept 7

A distributor had cost of goods sold of \$48 million last year and average inventory of \$6 million. ~~Sales were \$60 million. It moved into a new 300,000-square-foot facility.~~ What are its inventory turns and days of supply (365-day year)?

Target word: turns and days **Decides:** $48 / 6 = 8$ turns; $365 / 8$.

A	Correct	$48 / 6 = 8$; $365 / 8 \approx 45.6$.
B	Wrong stem number	Uses sales instead of COGS.
C	Confusion set	Confuses turns with days of supply.
D	Inverted operation	Divides inventory by COGS.

11. Answer B Concept 7 • transfer

A retailer targets 30 days of supply (use a 360-day year) and has annual COGS of \$36 million. What average inventory does that imply?

Target word: solve for inventory **Decides:** Turns = $360 / 30 = 12$.

A	Inverted operation	Multiplies COGS by 30.
B	Correct	$36M / 12 = 3.0M$.
C	Wrong stem number	The annual COGS.
D	Confusion set	Treats 30 days as 30 turns.

12. Answer D Concept 8

A clothing brand ~~with 400 stores and a new CEO~~ wants to cut greenhouse gas emissions across its supply chain. Most emissions come from contract factories that dye and sew its garments. Under the GHG Protocol, those factory emissions are:

Target word: scope **Decides:** The factories belong to suppliers.

A	Absolute language	Scope 3 reporting exists for exactly this.
B	Adjacent category	Scope 2 covers the company's own purchased electricity and heat.
C	Confusion set	Scope 1 covers sources the company owns or controls.
D	Correct	Value-chain emissions outside the company's own operations.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	5	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Bullwhip effect	Order swings grow as you move upstream; sharing real demand data and cutting lead times and batch sizes dampen them.	Adding buffers at every tier or ordering in bigger batches.	Small swings in consumer sales, large swings in upstream orders.
2	Push-pull boundary	Steps before the decoupling point run to forecast (push); steps after run to actual orders (pull).	Labeling a whole chain push or pull.	Some work done ahead, the rest after the customer chooses.
3	Make vs. buy on relevant cost	Compare only costs that differ: avoidable in-house costs vs. the purchase price plus supplier-management costs.	Comparing unit prices only or dropping costs that go away.	Variable cost, avoidable fixed cost, and an outside quote.
4	Risk as expected loss	Expected loss = probability × impact; rank risks on that product.	Ranking by probability or impact alone.	Two risks, each with a probability and a dollar impact.
5	Sales and operations planning	S&OP is a monthly cross-functional process that reconciles demand, supply, and finance into one agreed plan.	Letting each function keep its own number or confusing S&OP with scheduling.	Sales, operations, and finance each with a different number.
6	Supplier segmentation (Kraljic)	High-spend, high-supply-risk items are strategic and call for close partnerships; high-spend, low-risk items are leverage items for competitive bidding.	Bidding out a strategic item.	Few qualified suppliers plus critical and large spend.
7	Inventory turns and days of supply	Turns = COGS / average inventory; days of supply = days in the year / turns.	Using sales instead of COGS.	COGS, sales, and average inventory in one stem.
8	Scope 3 emissions	Emissions from suppliers and other value-chain partners are Scope 3; own operations are Scope 1; purchased energy is Scope 2.	Calling contract factories' emissions Scope 1.	Emissions at facilities the company does not own.

Course scope (public description): Global Supply Chain Management introduces a broad range of supply chain management concepts. Students consider supply chains and supply chain management strategies, including planning and design, risk management, and global and sustainable supply chain networks and management. Students learn about managing important relationships and their interdependencies. Students also investigate how supply chain costs affect consumers, quality and continuous improvement, and the role of the sales and operations planning (S&OP) process. Portions of this course are aligned with aspects of ASCM APICS certification programs, namely the CSCP and the CPIM.